TIMBOLIER INDUSTRIES, INC.

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DEW Line ™

Fuel Management Module Specifications

General Description

DEWLine Fuel Management Module is a system designed to aid a the vessel pilot in optimizing efficiency and tracking fuel usage. FMM connects to various engine and vessel sensors to compile data and display to the pilot fuel consumption and efficiency. Fuel readings are published on an NMEA2000 network for display on popular Multi-Function Displays (MFDs). DewLine FMM also has an iPhone App that displays fuel information on a phone or tablet.

System Components: Following is a complete list of hardware components in the DEWLine system. This specification is focused only on the FMM, which is highlighted below.

- ☐ Volumetric Fuel Flow Meters – measure fuel consumption by each engine, adjusted for temperature
- ☐ Fuel Management Module (FMM) that converts data from analog sensors and publish the data on the NMEA2k network.
- ☐ NMEA2000 (N2K) Network Backbone
- ☐ Multi-Function Display (MFD) – any of the popular marine navigations systems such as Garmin, SIMRAD, or Raymarine that display data gleaned from the N2K network
- ☐ iPhone App for display and processing of fuel data on a tablet or phone

FMM DESIGN SPECIFICATION

DEVELOPMENT ROADMAP: Initial board design must support all product generations.

VERSION 1: One FMM supports two engines. Generators not supported. Generator fuel consumption is a static input on the iPhone App for range (distance to empty) only. Limit 2 engines per vessel.

VERSION 2: One FMM supports two engines OR two generators. Generator fuel consumption measured and added to trip and range calculations. Up to 4 engines plus two generators per vessel using 3 FMM systems.

VERSION 3: Adds support for RPM data and cloud storage of data. Onboard storage is required for asynchronous cloud backup.

Chassis Requirements

- IP65 (or better) chassis such as the Polycase X64
- Waterproof connectors
- All external connectors must be on one side of the case.
- Flange with screw holes for mounting

Power

- DC 9v-30v input (supports low voltage condition on 12v systems up to 24v systems)
- Automatically boots when power applied.
- No user-initiated shut down is required prior to power off.
- If the FMM has a “sleep” mode, it will wake from sleep on sensor input activity.

Activity Lights

- Power light: On= Green, Sleep=yellow, Low voltage=red
- NMEA2000 Network light: Connected=Solid Blue, Activity=flashing blue

Environmental

- Must be able to operate in an ambient temperature of 45deg Celcius without overheating.

- Must be resistant to vibration to be able to function in an engine room.

Interoperability

- NMEA2000 communications must conform to the NMEA2000 specification.
- Multiple FMM devices must be able to live on the same network with unique identification of all outputs.

Field Service

- There must be a means provided for a field technician to flash a new image onto the device to update it as needed for bug fixes and feature enhancements.
- There must be a means for accessing data logs for troubleshooting purposes.
- Must be able to easily save and upload a configuration file either from the HMI or from a laptop using the USB port.

Network Connectivity

- NMEA200 port - Micro-C M12 5-pin screw connector
- USB 2.0 connector
- Wifi connectivity -- YES
- Ethernet - NONE

Table 1: Sensor Circuits

ID	Timbolier Config	Sensor Type	Range	Units	Signal	Signal Range	Supply Voltage	PGN	Fld	Src
1	Port Fuel Supply Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	Note 17	n/a	
2	Port Fuel Return Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	Note 17	n/a	
3	Stbd Fuel Supply Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	Note 17	n/a	
4	Stbd Fuel Return Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	Note 17	n/a	
5	Port Supply Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	130316	2,3,4	1,
6	Port Return Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	130316	2,3,4	2,
7	Stbd Supply Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	130316	2,3,4	1,
8	Stbd Return Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	130316	2,3,4	2,
9	Fuel tank lvl (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	127505	2,3	0x00
10	Fuel tank lvl (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	127505	2,3	0x00

Configuration Requirements

The following configuration parameters should be accessible via an android app that runs on the HMI. The configuration app will send NMEA2000 messages to the FMM to make configuration changes. The configuration app must also be able to "see" the current settings on the FMM so the user can see the current configuration using the app and determine what changes need to be made.

1. FMM Identification
    a. Specify if which engine or generator is associated with the FMM
        i. Port Drive Engine 1 or 2
        ii. Starboard Drive Engine 1 or 2
        iii. Generator 1, 2, 3, or 4
2. All Circuits
    a. Assign PGN for generator or drive entgine
    b. Turn off publication of PGNs because circuit not being used

3. Pulse Type Sensors
    a. Select NPN or PNP
    b. Assign factor of pulse frequency to ouput unit
    c. For Fuel Flow Sensors only:

i. Assign base temperature of fuel
ii. Assign fuel volume expansion coefficient due to temperature (typically .00095 for diesel fuel)
iii. All fuel flow calculations are normalized to the base temperature based on the current temperature reading at the fuel flow sensor.

4. Voltage Sensors
    a. Assign range: 0 ~ 5v; .5 ~ 5v; 0 ~ 12v; or other as needed
5. Resistance Sensors - Tankage
    a. Assign Resistance Range: High and Low (typically 240 – 33ohm or 0 – 190ohm)
    b.
6. Specify FMM identity relative to power unit:
    a. Port Engine
    b. Starboard Engine
    c. Generator 1
    d. Generator 2
7. For tank sensors
    a. input calibration data points at 25%, 50%, 75%, 100% of range, specifying gallons remaining.
    b. Set tank size
8. Trip Reset for fuel management (Trip 1, Trip 2, Trip 3) issued from App or MFD via NMEA2000.
9. For Fuel Flow sensors, must be able to input calibration value in milliliters. Default is 1.42ml. Range for calibration is 0 ~ 9.999ml.
10. NMEA2000 Inputs. The FMM will receive the following inputs via N2k:
    a. Trip fuel reset: 1,2,& 3.
    b. Distance to Waypoint if available on N2k Network
    c. Vessel Speed if available on N2k network - select speed over ground or speed over water

Table 2: Fuel Management Calculations

ID	Description	Calculation	Approximate Range	Units	PGN	Field/ Source	Freq (ms)
1	Instant Fuel Consumption (one for each engine)	Supply flow cycles less Return flow cycles over prior 3000ms for one engine only times volume (1.42) times 1800 (sec / hr)	0 – 3.2764	Cu M/hr	127497	5	1500
2	Trip 1 Fuel Consumption (one for each engine)	Accumulated Supply flow cycles less accumulated Return flow cycles for one engine since last Trip 1 reset times volume (1.42ml)	0 – 65.5320	Cu M	127497	2	3000
3	Trip 1 Run time	Accumulated time where Fuel Input is greater than X cycle per second. (x=greater than primer rate)	0 - 9,999,999 Seconds	Hrs:Min: Seconds	custom		3000
4	Trip 2 Fuel Consumption (one for each engine)	Accumulated Supply flow cycles less accumulated Return flow cycles for one engine since last Trip 2 reset times volume (1.42ml)	0 – 65.5320	Cu M	custom		3000
5	Trip 2 Run time	Accumulated time where Fuel Input is greater than X cycle per second. (x=greater than primer rate)	0 - 9,999,999 Seconds	Hrs:Min: Seconds	custom		3000
6	Trip 3 Fuel Consumption (one for each engine)	Accumulated Supply flow cycles less accumulated Return flow cycles for one engine since last Trip 3 reset times volume (1.42ml)	0 – 65.5320	Cu M	custom		3000
7	Trip 3 Run time	Accumulated time where Fuel Input is greater than X cycle per second. (x=greater than primer rate)	0 - 9,999,999 Seconds	Hrs:Min: Seconds	custom		3000

Reliability: The FMM provides critical data to the vessel pilot. Failure of the device could result in severe damage to the vessel and/or loss of life.

1. It must be completely stable and never require a re-boot.
2. Processing power must be sufficient to publish data on all 32 circuts simultaneously on a continuous basis for days at a time without rest or rebooting.

FUEL MANAGEMENT IPHONE APP

USER INPUTS

- Select SAE or Metric Display Values
- Select GPS speed from phone or GPS speed from FMM
- Name each Trip (1,2,3) e.g. "Ft. Laud to Grand Bahama"
- Add X gallons per hour for generator use when calculating Range

SYSTEM INPUTS

- Instant Fuel Use from FMM - each engine
- Cumulative Fuel Use from FMM - trip 1, 2, 3 - each engine
- Tank Level readings from FMM or N2K
- Speed readings from FMM or N2k

DISPLAY OUTPUTS

PORT ENGINE	**VESSEL**	**STBD ENGINE**	**Genset A**	**Genset B**
Instant Gal/Hr	Instant Speed (#2)	Instant Gal/Hr	Instant Gal/Hr	Instant Gal/Hr
	Instant combined Gal/Hr			
	Instant Gal/Mile (#1)			
	Range (Miles to Empty)			
Trip 1,2,3 Gal Used	Trip 1,2,3 combined Gal Used	Trip 1,2,3 Gal Used	Trip 1,2,3 Gal Used	Trip 1,2,3 Gal Used
	Trip 1,2,3 Efficiency (Gal/Mile) (#1)			
	Trip 1,2,3 Distance			
	Trip 1,2,3 Avg Speed			
	Trip 1,2,3 Run Time			
	Tank 1,2 fuel remaining (gal)			

Note #1 - Do not include generator fuel consumption in this calculation.
Note #2 - "Instant" values should be calculated based on the prior 3 seconds of data.

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Sensor Manufacturer/Part Number

ID	**Timbolier Config**	**Sensor Type**	**Range**	**Units**	**Signal**	**Signal Range**	**Supply Voltage**	Manu-facturer	
1	Engine Tach (RPM)	Pulse	0-8000	RPM	PWM	0-100 Hz	??	Glen-denning	Synchronizer output
2	Engine Load	Д volts	0-5	%	volts	1kHz	5V	Vishay	981HEB1WCF
3	Fuel Supply Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	MiSen-sor	FM-006
4	Fuel Return Rate	Pulse	.05 – 16	L/min	NPN	0 ~ 200 Hz	+9~24v	MiSen-sor	FM-006
5	Speed over water	Pulse	0 - 40	m/sec	Pulse	TBD	??	Law-rance	99-75
6	Engine Oil Pressure	Д volts	0 - 500	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-5bV1P1N1W3T1
7	Gear Oil Pressure	Д volts	0 - 1700	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-17bV1P1N1W3T1
8	Fuel Pressure	Д volts	0 - 500	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-5bV1P1N1W3T1
9	Crankcase Pressure	Д volts	-2 to +5	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-K5V1P1N2W3T1
10	Turbo Boost	Д volts	0-250	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-2.5bV1P1N2W3T1
11	Raw Water pressure A	Д volts	0-100	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-1bV1P1N2W3T1
12	Raw Water pressure B	Д volts	0-100	kPa	volts	.5 – 4.5v	5v	Yunyi	P140-1bV1P1N2W3T1
13	Supply Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	MiSen-	FM-006
14	Return Fuel Temp	Д volts	-20 ~ +80	Deg C	volts	.3 -- 1.3v	5v	MiSen-	FM-006
15	Starting Bank Volts	sense Volts	0 -- 25	Volts	volts	n/a	n/a	n/a	n/a
16	House Bank Volts	sense volts	0 – 25	Volts	volts	n/a	n/a	n/a	n/a
17	Engine Charging Amps	Shunt	0 – 150	Amps	volts	50 / 100mv?	n/a	Reidon	MKA-150 50mV
18	House Charging Amps	Shunt	0 – 150	Amps	volts	50 / 100mv?	n/a	Reidon	MKA-150 50mV
19	Exhaust Gas Temp A	T/C	0 - 800	Deg C	mV	Type K	n/a	any	any type k
20	Exhaust Gas Temp B	T/C	0 - 800	Deg C	mV	Type K	n/a	any	any type k
21	Engine Coolant Temp	Д Ohms	100 - 240	Deg F	Ohms	410~30 Ω	12v	Sierra	TS26631
22	Engine Oil Temp	Д Ohms	100 - 300	Deg F	Ohms	1050~26 Ω	12v	Murphy	10702010
23	Turbo #1 Temp	Д Ohms	100 - 300	Deg F	Ohms	1050~26 Ω	12v	Murphy	10702010
24	Turbo #2 Temp	Д Ohms	100 - 300	Deg F	Ohms	1050~26 Ω	12v	Murphy	10702010
25	Gear Oil Temp	Д Ohms	100 - 240	Deg F	Ohms	410~30 Ω	12v	Sierra	TS26631
26	Fuel tank lvl (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	any	std float sensor
27	Water Tank lvl (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	any	std float sensor
28	Waste Tank lvl (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	any	std float sensor
29	Livewell Tank (note 3)	Д Ohms	0 – 100	PCT (%)	Ohms	240~33 Ω	12v	any	std float sensor
30	Water in Fuel Switch	Open/Close	n/a	n/a	n/a	Note 1	12v	Entra-tech	W004811AA
31	Low Coolant Switch	Open/Close	n/a	n/a	n/a	n/a	12v	Kysor	KYS 5022 01185 01
32	High Bilge water lvl	Open/Close	n/a	n/a	n/a	n/a	12v	any	float switch on/off